

 FIAT CHRYSLER AUTOMOBILES	HEXAGON HEAD SCREW AND CONED WASHERS - MATPOINT		SP-A0262
			Standard Parts
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		Date: 10-DEC-2019	
# 06104422AA REF. # 68305239AA ASF CONTROLLED WHEN RED ASF Engineering Approved 5/13/24 10 Pages			
Change level	Date	Description of change	
-	12-OCT-2016	Initial release	
A	05-FEB-2019	Updated part number table and ref standards	
B	10-DEC-2019	ADDED NEW PART 06104245AA	
For information, please contact author and co-author in the lateral label of this document. For APAC, LATAM and Maserati contacts, please consult SD.00130			
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HEXAGON HEAD SCREW AND CONED WASHERS - MATPOINT		
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1 GENERAL 1.1 Purpose All parts in this standard are released as Change (PCL) A. The document change control follows normal corporate change increments. 1.2 Coverage of this Standard Standard Parts shall be identified by the category code: 014.010 (Hexagon Head Screw and Coned Washers) 1.3 Limitations on Usage Not applicable 2 REFERENCES			
Table 1 - References			
Document Number	Shield/Designator (if applicable)	Document Title	downloadable for suppliers from beSTandard
PS-11036		CORROSION RESISTANT COATING - ZINC-RICH / ALUMINUM-RICH - CHROMIUM-FREE- NON-ELECTROLYTIC	Y
PS-10378		CORROSION RESISTANT COATING - ZINC-RICH / ALUMINUM-RICH - NON-ELECTROLYTIC - SILVER OR BLACK MULTI-COAT SYSTEM	Y
01307		REPRESENTATION, DESIGNATION AND MARKING FOR THREADS, SCREWS AND NUTS - STEEL BOLTS AND NUTS	Y
MS.50077	<S>	EXTERNALLY THREADED FASTENERS FOR NORMAL USE	Y
PS.50023	<S>	HYDROGEN EMBRITTLEMENT RELIEF	Y
CS.00031		MATPOINT DIMENSIONAL AND USAGE REQUIREMENTS	Y
PS-7099		FASTENER GENERAL SPECIFICATIONS	Y
3 DEFINITIONS/ABBREVIATIONS/ACRONYMS/SYMBOLS SP STANDARD PARTS CATEGORY NUMERICAL CODE THAT PRE DEFINES A NOUN NAME OF A FASTENER CODE TYPE ISO INTERNATIONAL STANDARDS ORGANIZATION FCA FIAT CHRYSLER AUTOMOBILES			
4 CORRELATION TO OTHER STANDARDS Not applicable			
5 REGULATED SUBSTANCES & RECIBLABILITY Not applicable			

6 GEOMETRY

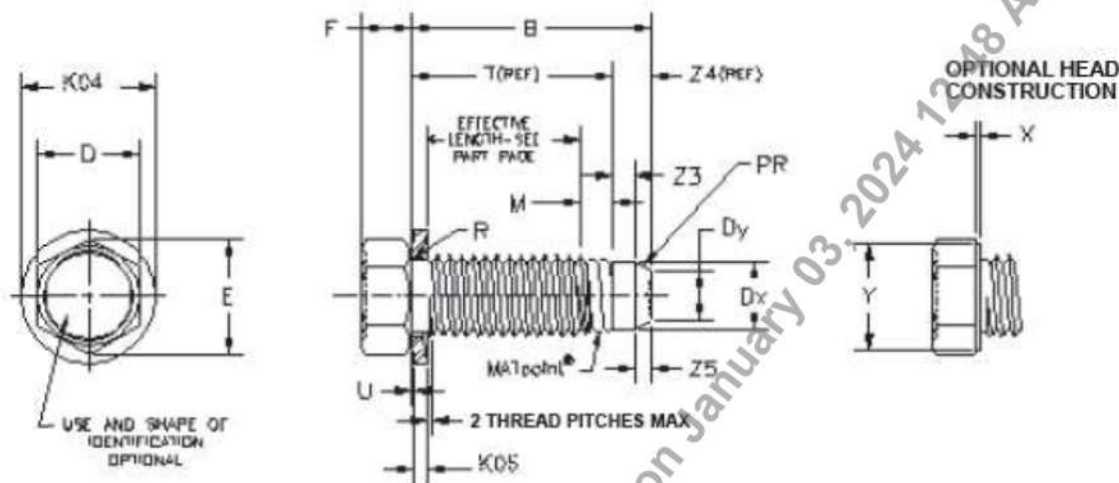


Figure 1 - Hexagon Head Screw & coned washer – Matpoint

Table 2 - Dimensional requirements

Nom Size	Thd Pitch	B		D		E	K04		K05		U		R		X		Y		
		Head height		Head width		Width-corners	Washer OD		Washer thickness		Washer crown hgt		Fillet radius		Washer face thk.		Washer face dia	Panel Hole dia.	Panel Slot wdt.
		Min	Max	Min	Max	Min	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Max
M5	0.8	3.35	3.65	7.78	8.00	8.71	16.3	17.0	1.54	1.80	0.40	0.65	0.20	0.35	0.2	0.5	7.2	14.5	6.5
M6	1.00	4.13	4.47	9.76	10.0	10.93	19.16	20.0	2.06	2.32	0.50	0.80	0.25	0.40	0.2	0.5	9.0	16.0	10.0
M8	1.25	5.10	5.50	12.73	13.0	14.26	19.16	20.0	2.37	2.63	0.50	0.80	0.40	0.60	0.3	0.6	11.7	17.0	11.0
M10	1.5	6.17	6.63	15.57	16.0	17.59	23.16	24.0	2.74	3.00	0.60	0.90	0.40	0.60	0.3	0.6	13.5	19.0	13.0
M12	1.5	7.24	7.76	17.67	18.0	19.79	31.00	32.00	3.25	3.51	0.80	1.15	0.60	1.10	0.3	0.6	16.2	27.0	18.0
M14	1.5	8.51	9.09	20.64	21.0	23.12	36.00	37.00	3.80	4.06	1.00	1.30	0.60	1.10	0.3	0.6	18.9	31.0	21.0

7 TABLE OF PART NUMBERS

Table 3 - Thread Sizes and Finishes with Matpoint

Nominal Size	Pitch	Washer Diameter	Length	Finish	Strength (Property class)	Part Numbers
M6	1.00	20.0	16	PS-11036	8.8	06104364AA
M6	1.00	20.0	20	PS-11036	8.8	06104366AA
M6	1.00	20.0	25	PS-11036	8.8	06104368AA
M6	1.00	20.0	30	PS-11036	8.8	06104370AA
M6	1.00	20.0	35	PS-11036	8.8	06104372AA
M6	1.00	20.0	40	PS-11036	8.8	06104374AA
M6	1.00	20.0	45	PS-11036	8.8	06104376AA
M6	1.00	20.0	50	PS-11036	8.8	06104378AA
M6	1.00	20.0	55	PS-11036	8.8	06104380AA
M6	1.00	20.0	60	PS-11036	8.8	06104382AA
M8	1.25	20.0	20	PS-11036	8.8	06104384AA



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Table 3 - Thread Sizes and Finishes with Matpoint

Nominal Size	Pitch	Washer Diameter	Length	Finish	Strength (Property class)	Part Numbers
M8	1.25	20.0	25	PS-11036	8.8	06104386AA
M8	1.25	20.0	30	PS-11036	8.8	06104388AA
M8	1.25	20.0	35	PS-11036	8.8	06104390AA
M8	1.25	20.0	40	PS-11036	8.8	06104392AA
M8	1.25	20.0	45	PS-11036	8.8	06104394AA
M8	1.25	20.0	50	PS-11036	8.8	06104396AA
M8	1.25	20.0	55	PS-11036	8.8	06104398AA
M8	1.25	20.0	60	PS-11036	8.8	06104400AA
M8	1.25	20.0	65	PS-11036	8.8	06104402AA
M8	1.25	20.0	70	PS-11036	8.8	06104404AA
M8	1.25	20.0	80	PS-11036	8.8	06104406AA
M8	1.25	20.0	90	PS-11036	8.8	06104408AA
M8	1.25	20.0	100	PS-11036	8.8	06104410AA
M10	1.50	24.0	25	PS-11036	10.9	06104412AA
M10	1.50	24.0	30	PS-11036	10.9	06104414AA
M10	1.50	24.0	35	PS-11036	10.9	06104416AA
M10	1.50	24.0	40	PS-11036	10.9	06104418AA
M10	1.50	24.0	45	PS-11036	10.9	06104420AA
M10	1.50	24.0	50	PS-11036	10.9	06104422AA
M10	1.50	24.0	55	PS-11036	10.9	06104424AA
M10	1.50	24.0	60	PS-11036	10.9	06104426AA
M10	1.50	24.0	65	PS-11036	10.9	06104428AA
M10	1.50	24.0	70	PS-11036	10.9	06104430AA
M10	1.50	24.0	80	PS-11036	10.9	06104432AA
M10	1.50	24.0	90	PS-11036	10.9	06104434AA
M10	1.50	24.0	100	PS-11036	10.9	06104436AA
M12	1.50	32.0	25	PS-11036	10.9	06104438AA
M12	1.50	32.0	30	PS-11036	10.9	06104440AA
M12	1.50	32.0	35	PS-11036	10.9	06104442AA
M12	1.50	32.0	40	PS-11036	10.9	06104444AA
M12	1.50	32.0	45	PS-11036	10.9	06104446AA
M12	1.50	32.0	50	PS-11036	10.9	06104448AA
M12	1.50	32.0	55	PS-11036	10.9	06104450AA
M12	1.50	32.0	60	PS-11036	10.9	06104452AA
M12	1.50	32.0	65	PS-11036	10.9	06104454AA
M12	1.50	32.0	70	PS-11036	10.9	06104456AA
M12	1.50	32.0	80	PS-11036	10.9	06104458AA
M12	1.50	32.0	90	PS-11036	10.9	06104460AA
M12	1.50	32.0	100	PS-11036	10.9	06104462AA
M12	1.50	32.0	110	PS-11036	10.9	06105106AA
M12	1.50	32.0	120	PS-11036	10.9	06105107AA
M12	1.50	32.0	130	PS-11036	10.9	06105108AA
M12	1.50	32.0	140	PS-11036	10.9	06105109AA
M14	1.5	37.0	50	PS-11036	10.9	06105110AA
M14	1.5	37.0	55	PS-11036	10.9	06105111AA
M14	1.5	37.0	60	PS-11036	10.9	06105112AA
M14	1.5	37.0	70	PS-11036	10.9	06105113AA
M14	1.5	37.0	80	PS-11036	10.9	06105114AA
M14	1.5	37.0	90	PS-11036	10.9	06105115AA
M14	1.5	37.0	100	PS-11036	10.9	06105116AA
M14	1.5	37.0	110	PS-11036	10.9	06105117AA
M14	1.5	37.0	120	PS-11036	10.9	06105118AA
M14	1.5	37.0	130	PS-11036	10.9	06105119AA
M14	1.5	37.0	140	PS-11036	10.9	06105120AA
M14	1.5	37.0	145	PS-11036	10.9	06105245AA
M14	1.5	37.0	150	PS-11036	10.9	06105121AA



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Table 3 - Thread Sizes and Finishes with Matpoint						
Nominal Size	Pitch	Washer Diameter	Length	Finish	Strength (Property class)	Part Numbers
M14	1.5	37.0	160	PS-11036	10.9	06105122AA

Table 3 - Thread Sizes and Finishes with Matpoint						
Nominal Size	Pitch	Washer Diameter	Length	Finish	Strength (Property class)	Part Numbers
M6	1.00	20.0	20	PS-10378 Black	8.8	06105218AA
M6	1.00	20.0	25	PS-10378 Black	8.8	06105219AA
M6	1.00	20.0	30	PS-10378 Black	8.8	06105220AA
M8	1.25	20.0	25	PS-10378 Black	8.8	06105221AA
M8	1.25	20.0	30	PS-10378 Black	8.8	06105222AA
M8	1.25	20.0	35	PS-10378 Black	8.8	06105223AA
M8	1.25	20.0	40	PS-10378 Black	8.8	06105224AA
M8	1.25	20.0	45	PS-10378 Black	8.8	06105225AA
M8	1.25	20.0	50	PS-10378 Black	8.8	06105226AA
M10	1.50	24.0	25	PS-10378 Black	10.9	06105227AA
M10	1.50	24.0	30	PS-10378 Black	10.9	06105228AA
M10	1.50	24.0	35	PS-10378 Black	10.9	06105229AA
M10	1.50	24.0	40	PS-10378 Black	10.9	06105230AA
M10	1.50	24.0	45	PS-10378 Black	10.9	06105231AA
M10	1.50	24.0	50	PS-10378 Black	10.9	06105232AA
M12	1.50	32.0	25	PS-10378 Black	10.9	06105233AA
M12	1.50	32.0	30	PS-10378 Black	10.9	06105234AA
M12	1.50	32.0	35	PS-10378 Black	10.9	06105235AA
M12	1.50	32.0	40	PS-10378 Black	10.9	06105236AA
M12	1.50	32.0	45	PS-10378 Black	10.9	06105237AA
M12	1.50	32.0	50	PS-10378 Black	10.9	06105238AA

Application: The screw and coned washer assembly covered by this standard is for general high volume usage with pre-tapped holes and nuts.

8 GENERAL SPECIFICATION

- A. **Unspecified dimensions:** For detail pertaining to dimensions or unspecified specification, screws shall conform to the specification shown in PS-7099, unless otherwise noted.
- B. **Head:** Dimensions across flats and across corners of the head shall be measured at the point of maximum metal. Bearing surface of the head shall be at right angles to the axis of the screw within a tolerance of 2° and concentric with the axis of the screw within a tolerance of 3 percent of the maximum width across flats. Taper of sides of hex (angle between one side and the axis) shall not exceed 2° or 0.10 mm whichever is greater, the specified width across flats being the large



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dimension. A slight rounding of all edges and corners of the hex surfaces of indented hex head is permissible provided the diameter of the bearing circle is not less than 90 percent of the minimum "Across Flats" dimension.

The depth of indentation shall not exceed 25 percent of minimum head height.

- C. **Length:** The length of screw shall be measured parallel to the axis of the screw from the largest diameter of the bearing surface of the head to the extreme point. The tolerance on length for screws shall be as shown in Table 4.

Table 4 - Length Tolerance		
+/- Tolerance		
Over 6mm	Thru 10mm	0.29
10mm	18mm	0.35
18mm	30mm	0.42
30mm	50mm	0.50
50mm	80mm	0.60
80mm	120mm	0.70
120mm	180mm	0.80
180mm	250mm	0.93
250mm	315mm	1.05
315mm	400mm	1.15

- D. **Threads:** ISO Metric Screw Threads tolerance class 6g.
- E. **Straightness:** Shanks of bolts shall be straight within the following limits: for bolts with nominal lengths to and including 300mm, the maximum camber shall be 0.006 mm per mm of bolt length.
- F. **Defects:** Screw and Washer Assemblies must be free from all defects which might affect their serviceability.
- G. **Hydrogen embrittlement relief:** Parts shall conform to PS.50023<S>
- H. **Washer:** Coned washer must withstand load to flatten for 24 hours without cracking. Washer must be free to rotate after assembly with screw and be retained by the last thread. Sharp edges and burrs must be removed by tumbling before assembly with screw. Washers are designed to flatten at less than 70% of screw proof load.
- I. **Identification:** 01307
- J. **MATPOINT:** Matpoint shall conform to specification and dimension in CS.00031 and as shown in Table 5 and Figure 2

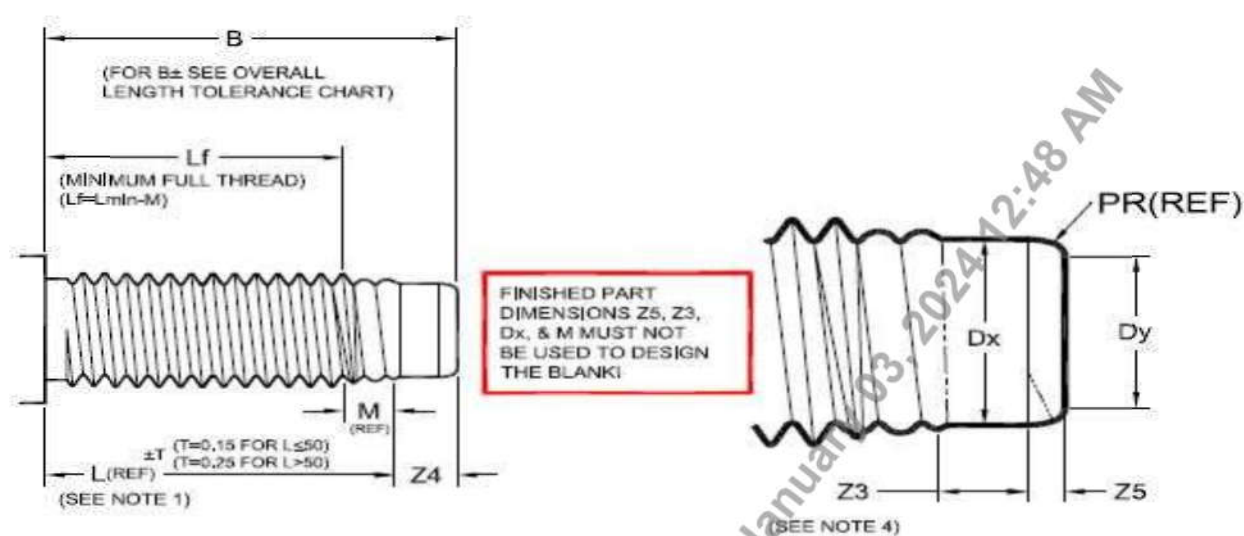


Figure 2 - Matpoint dimensional requirements

Table 5 - Matpoint® dimensional requirements

THREAD SIZE & PITCH	PR REF	Dy Max	W +/-0.1	Dx MAX	Dx MIN	Z3 MIN	Z4 Max	Z5 MIN	M REF
M5x0.8	1.5	3.4	0.16	4.03	3.987	1.65	2.5	0.6	1.8
M6x1.0	1.8	4.1	0.2	4.8	4.724	1.75	2.85	0.75	2.3
M8x1.25	2.3	5.4	0.25	6.54	6.447	2.49	3.9	1	2.8
M10x1.25	3.00	7.5	0.25	8.54	8.447	2.60	4.05	1.25	2.8
M10x1.5	2.8	6.8	0.3	8.23	8.143	3.15	4.65	1.25	3.4
M12x1.5	3.3	8.2	0.3	10.23	10.143	3.94	5.65	1.5	3.4
M12x1.75	3.3	8.2	0.35	9.95	9.88	3.15	5.65	1.5	4
M14x1.5	4.1	9.5	0.3	12.23	12.143	4.18	6.43	1.75	3.4
M14x2.0	4.1	9.6	0.4	11.72	11.604	3.68	6.43	1.75	4.5
M16x1.5	4.9	12.8	0.3	14.26	14.143	4.1	6.6	2	3.4
M16x2.0	4.7	10.9	0.4	13.72	13.609	4.1	7.1	2	4.5

K. **Material:** Steel – MS.50077 (8.8A and 10.9A property classes).

L. **Thread Length:**

For Fasteners Having a Nominal Length of 50mm or Shorter: Fasteners 50mm in length or shorter shall be threaded for the entire length of the screw. A full formed thread shall extend to a maximum distance of one thread pitch from the bearing face of the washer, or closer if practicable (see Figure 2).

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Figure 3 - For Fasteners Having a Nominal Length of 50mm or Shorter For Fasteners Having A Nominal Length Longer Than 50mm: Fasteners longer than 50mm in length shall be threaded for the entire length of the screw (see Paragraph I and Figure 3), or at the option of the manufacturer have a minimum of 45mm of full formed thread (see Figure 4). Figure 4 - For Fasteners Having A Nominal Length Longer Than 50mm Recommended Gaging Practice for Fasteners Threaded Full Length: The recommended method of measuring the unthreaded portion of the screw is to assemble the fastener into a non-counterbored or non-countersunk standard 6g thread ring gage of the appropriate size and class. The gage and screw shall be assembled by hand as far as the thread will permit using reasonable effort. The distance measured between the face of the gage and the bearing face of the washer shall not exceed one thread pitch (see Figure 5). Figure 5 - Recommended Gaging Practice for Fasteners Threaded Full Length 9 MATERIALS, FINISH AND SURFACE COATING Material specification shall be as designated and according to specification Steel MS.50077<S> (for property class 8.8A and 10.9A) Parts made from this standard shall be coated as per specification. - PS-11036 – Aluminum Rich dip spin coating		

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- PS-10378 (Black) - Zinc rich / aluminum rich Multi coat system 10 SPECIAL QUALITY REQUIREMENTS Not applicable 11 APPROVED SOURCE LIST Not applicable Printed by W97428D@auth.fcagroup.com on January 03, 2024 12:48 AM		